



Analysis of Patterns of Neck Trauma and its Outcomes at Level I Trauma Centre: An Ambispective Observational Study

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Abstract

Traumatic neck injuries (TNIs) constitute 5–10% of all trauma cases. Though the neck injuries are not frequent but the morbidity and mortality rates associated with these injuries are high. It was a single centre, ambispective observational study. One hundred fifty-six Patients with TNI of either sex > 14 years of age and giving an informed consent by self or their legally authorized representative getting admitted under trauma surgery were included in study. On comparing retrospective Vs prospective cohort penetrating injury was more common than blunt injury but there is increase in the rate of blunt injury i.e., from 3.2 to 17.7%. In retrospective cohort most common mechanism was assault (46.6%) followed by road traffic injury (26%). While in the prospective cohort the most common mechanism was road traffic injury (22.6%) followed by assault (22%). PNI (Penetrating neck injury) was the most common injury pattern predominantly following assault or self-inflicted harm. But in the recent times there was a change in the trend toward blunt neck injury pattern.

Keywords Traumatic neck injury · Patterns · Observational study · Outcome

Introduction

Traumatic injury is a major health problem affecting mainly the young productive youth worldwide. Trauma related morbidity and mortality is still high (1). Traumatic neck injuries (TNIs) constitute 5%–10% of all trauma cases (1). Though the neck injuries are not frequent but the morbidity and mortality rates associated with these injuries are high (2). They can be potentially life-threatening due to the involvement of the aerodigestive tract and major neck vessels that are relatively unprotected. Hence a timely resuscitation and intervention with a multidisciplinary team approach is critical for the successful outcome of these patients.

Literatures have broadly classified TNI into two groups i.e., penetrating and blunt neck injury primarily based on the mechanism of the injury. Although the incidence of penetrating neck injuries predominates over the blunt, but in recent decades, the incidence of injuries following blunt impact has increased, primarily attributed due to the road traffic injuries (3, 4).

Management of traumatic neck injuries pose a surgical challenge because multiple vital structures are vulnerable to injuries in the small, confined unprotected area (5). Priorities for treating patients with neck injuries are the same as for any other acutely injured patients following trauma primarily based on the standard trauma management protocol i.e., the Advanced Trauma life support (ATLS) of the American College of Surgeons. Majority of the literatures on TNI are from western developed world and only a few literatures if any exists, addresses the issues in the low- and middle-income countries, inhabited by more than two third of the world's population.

Materials and Methods

It was a single centre, ambispective observational study. Patients with TNI of either sex > 14 years of age and giving an informed consent by self or their legally authorized representative getting admitted under trauma surgery were included in study. Considering the study by Hassan Al-Thani et al. (2015), they have reported a proportion of traumatic neck injuries approximately 5–10% of all severe trauma cases. So, for 10% traumatic neck injuries at 95% confidence level with 5% allowable margins of errors, we need to study

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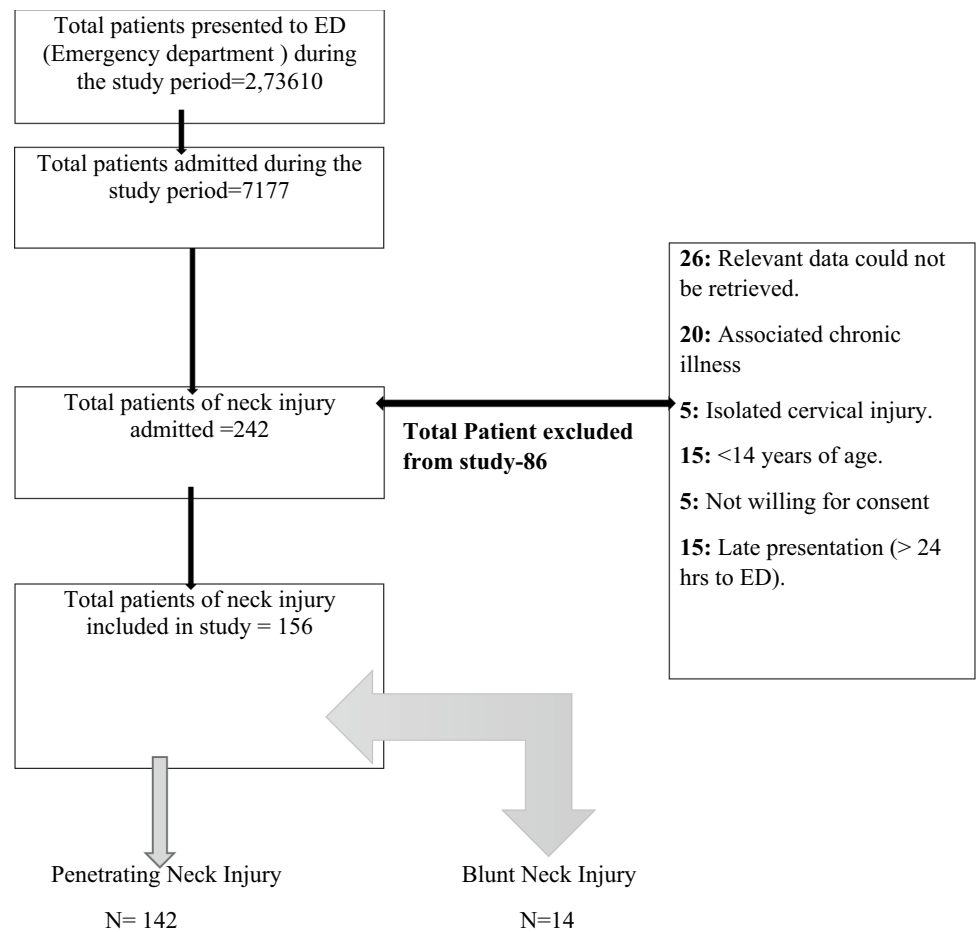
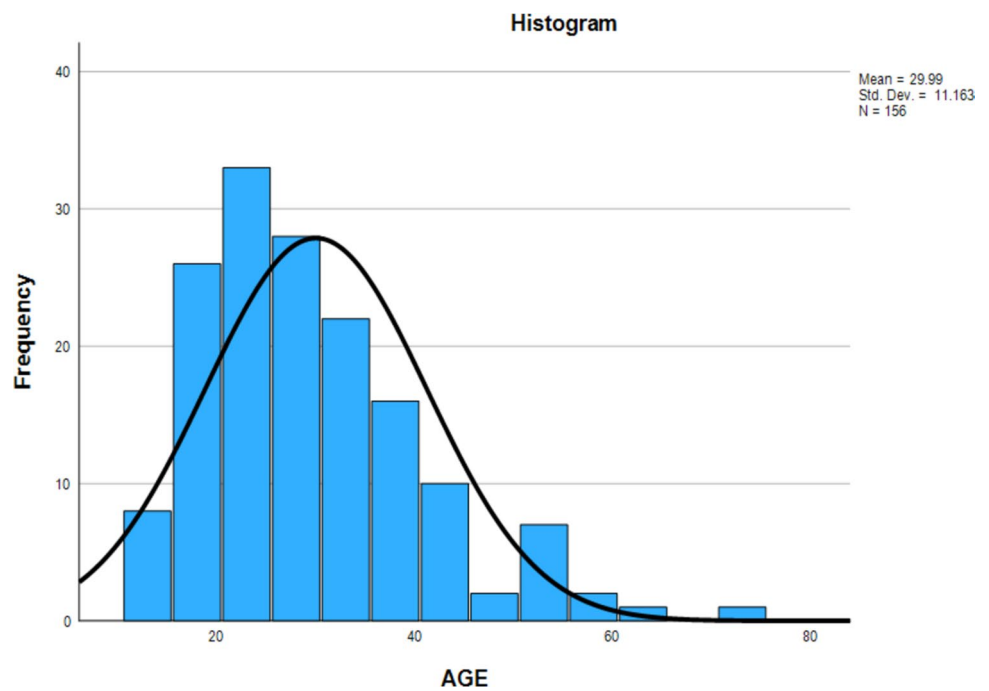
Fig. 1 Histogram of Age**Fig. 2** Flow diagram of the study

Table 1 Comparison between retrospective Vs prospective cohort

	Retrospective cohort		Prospective cohort	
	Number of patients (n)	Percentage (%)	Number of patients(n)	Percentage (%)
Number	94	100	62	100
Age(years) Mean (SD)	29.5(10.3)		30.82(11.8)	
<i>Sex</i>				
Male	86	91.5	56	90.3
Female	08	8.5	06	9.7
<i>Mechanism of injury</i>				
Penetrating neck injury	91	96.8	51	82.3
Blunt neck injury	03	3.2	11	17.7
<i>Spectrum of mechanism of injury</i>				
Assault	40	46.6	14	22.6
Self-inflicted injury	19	22.3	14	22.6
Road traffic injury	21	26.3	17	27.4
Accidental injury	11	11.7	08	12.9
Gunshot injury	01	1.1	03	4.8
Manja Injury	02	2.1	02	3.21
Bull gore injury	00	00	01	1.6
Blast Injury	00	00	01	1.6
Wooden branch related injury	00	00	01	1.6

Table 2 Spectrum of the mechanism of neck injury

Mechanism of neck injury		n (%)		Number of patients (n-156)	Percentage (%)
Penetrating neck injury	Low-velocity	138 (88.4)	Assault	47	30.1
			Self-inflicted injury	36	23.0
			RTI	32	20.5
			Accidental	16	9.6
			Manja Injury	04	2.5
			Bull Gore injury	01	0.6
			Blast Injury	01	0.6
			Wooden branch injury	01	0.6
			Gunshot Injury	04	2.5
			Assault	04	2.5
Blunt neck injury	High-velocity	4(2.5)	RTI	06	3.8
			Accidental	04	2.5

Table 3 Mean age

Age (in years)	Mean	Median	Mode	SD
	30	28.00	25	11.16

approximately 144 samples. So, the total sample calculated for study was 144 (Fig. 1).

To describe the patient's characteristics (demographics, injury details, clinical parameters, outcome) descriptive statistics were used. Quantitative data were presented

as Mean $\pm$ SD if normality of the variable was established otherwise Median (min, max). Data were tested for normality using the Kolmogorov–Smirnov test. Student's t-test was used to compare the normal variables, whereas the Mann–Whitney U-test was performed to compare the non-normal variables. For comparison of categorical data, the Chi-square/Fisher's exact test/ Fisher-Freeman-Halton exact test was used. For comparisons between categorical and non-categorical data for > 2 groups, the one-way ANOVA test was used. All the data were summarized and analysed with the used of Statistical Package for Social Sciences (SPSS,

Table 4 Anatomical zone of the neck

		Number of patients (n-156)	Percentage (%)
Anatomical zone of neck	Zone I	21	13.5
	Zone II	126	80.8
	Zone III	03	1.9
	Multiple Zone	06	3.8

version 29.0 software. A value of $p < 0.05$ was considered statically significant.

Results

A total of 156 patients with TNI were recruited. In the retrospective cohort, total of 94 patients and in the prospective cohort total of 62 patients were recruited for the study. Penetrating injury was the predominant mechanism of the neck injury seen in 91.0% ($n = 142$) patients while 9.0% had blunt injury. In penetrating injury majority were low velocity injuries made up to 97.3% of all neck injuries and in that most common mechanism of neck injury was an assault in 51 patients (32.7%) followed by in decreasing order RTI in 38 (24.4%), Self-inflicted injury in 36 (21.1%), accidental injury like fall over sharp object or fall of sharp object and dupatta injury in 19(12.2%), Manja injury in 4(2.6%). Bull gore, blast injury and wooden branch related injury 1(0.6%) patients in each group. High velocity penetrating injury was least common.

Table 5 Anatomical distribution of traumatic neck injury

Anatomical distribution of traumatic neck injury	n (%)		Number of patients (n-152)	Percentage (%)
Only soft tissue neck injury	90(58.8)	Superficial tissue of neck	23	15
		Deep tissue of neck	67	43.7
Aerodigestive injury	34(22.2)	Laryngeal Injury	17	11.11
		Tracheal Injury	15	9.80
		Pharyngeal injury	01	0.6
		Esophageal Injury	01	0.6
Vascular Injury	24(15.6)	Carotid Artery (CA)	05	3.2
		Subclavian artery (SCA)	05	3.2
		Internal jugular vein (IJV)	10	6.4
		External jugular vein (EJV)	02	1.3
		Minor vessels	02	1.3
		Brachial plexus injury (BPI)	02	1.3
Nerve injury	02(1.3)	Thoracic duct injury (TDI)	01	0.6
Thoracic duct injury	01(0.6)	Parotid gland Injury	01	0.6
Miscellaneous	01(0.6)			
Missing	04			

Table 6 In-hospital mortality

	Number of patients (n = 156)	Percentage (%)
Discharge	152	97.4
In-hospital mortality	04	2.6
Early Mortality	00	00
Late Mortality	04	2.6
Cause of mortality		
Haemorrhagic shock	01	0.6(True mortality)
Traumatic brain injury	01	0.6
Septic shock	01	0.6
Crush syndrome	01	0.6
Total	156	100

The common mechanism of injury observed in penetrating injury was assault (33.09%) and in blunt injury was RTI (42.9%). Zone II was common in both groups. Zone III injury was not observed in blunt injury.

On comparing retrospective Vs prospective cohort there were no difference observed in gender and the mean age. Penetrating injury was more common than blunt injury but there is increase in the rate of blunt injury i.e., from 3.2 to 17.7%. In retrospective cohort most common mechanism was assault (46.6%) followed by road traffic injury (26%). While in the prospective cohort, road traffic injury (22.6%) observed slight higher frequency then assault (22%).

In our study, we observed blunt neck injury was noted in about 9% of all traumatic neck injuries and in 43% cases it was following RTI. This change in mechanism could be attributed to covid era with restrictions at public gathering and the increased number of trauma cases following MVC (Motor vehicle collision) due to RTI (Fig. 2 and Tables 1, 2, 3).

Spectrum of mechanism	Blunt neck trauma
Assault	04(28.6)
Road traffic injury	06(42.9)
Accidental injury	04(28.6)

Degree of Penetration Assessment

Degree of penetration assessed by anatomical zone of the neck injury and anatomical distribution of neck injury and in hospital mortality (Tables 4, 5, 6).

Discussion

In our study we observed 3.3% prevalence of Traumatic neck injuries. total 156 studied, 142 (91.0%) patients were males, with a male-to-female ratio of 10:1. The mean age of the patients were 29.9 and SD of ± 11.16 years.

Table 7 Comparison between various study on traumatic neck injury

Author	Country/Region	Study Period (Years)	Study type	Size (n)	Mean AGE	SEX (M) (%)	MOI (%)	Mortality (%)
Present study	India	5.5	Amibi-spective	156	29.9	91	Assault-37.2 RTI -24.7 Self-inflicted injury -21.1 Accidental-12.2 Gunshot-2.6	2.6
Al-Thani et al	Qatar		Retro-spective	51	31	98	MVC-29.4 Stabbing-17.6 Machinary-17.6 Fall from hight-9.8	11.8
Forner et al	America	5	Retro-spective	20.285	BNI-45 PNI-35.8	64.5	–	5
Zakaria et al	UK	5	Retro-spective	31	36	87.4	Assault-55.2 Self-inflicted injury-38%	1.29
Ajiya et al	Nigeria	7	Retro-spective	35	36.7	88.6	Arrow injury-25.7 Self-inflicted injury-20 Gunshot -11.4	2.9
Irish et al	Canada	10	Retro-spective	85	34.5	91	BNI-MVC-53 Hockey -21 PNI-MVC-65 Gunshot -8	3
Gliyoma et al	Tanzania	4	Retro-spective	98	26	70.4	Assault-55 Self-inflicted injury-34.7 RTI-10.2	11.2

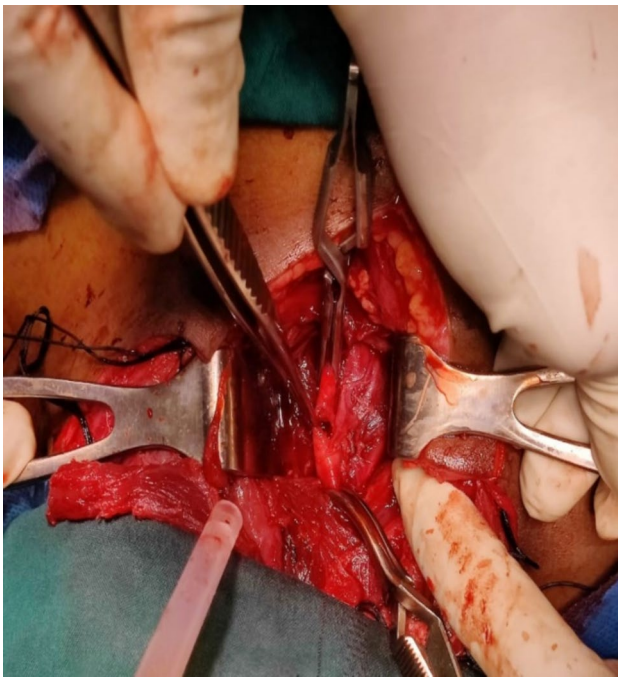


Fig. 3 Through and through carotid injury



Fig. 5 Platysma Breach (Zone II)

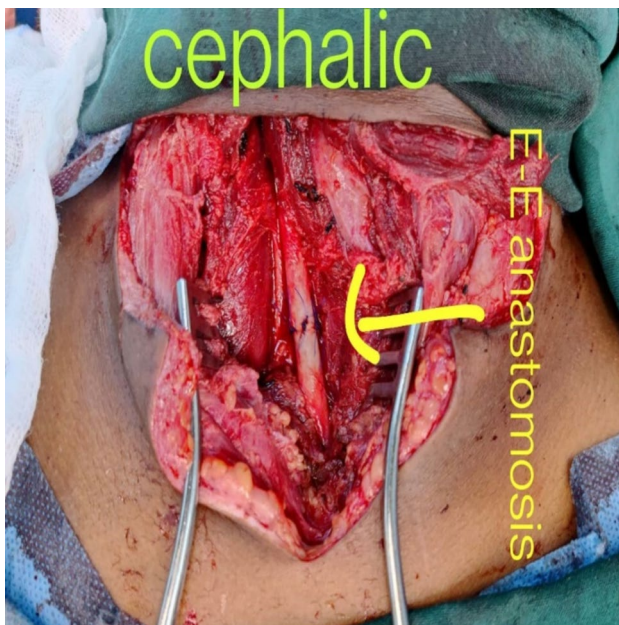


Fig. 4 End to end anastomosis of carotid artery



Fig. 6 Stab injury (Zone I)



Fig. 7 Neck injury in multiple zones

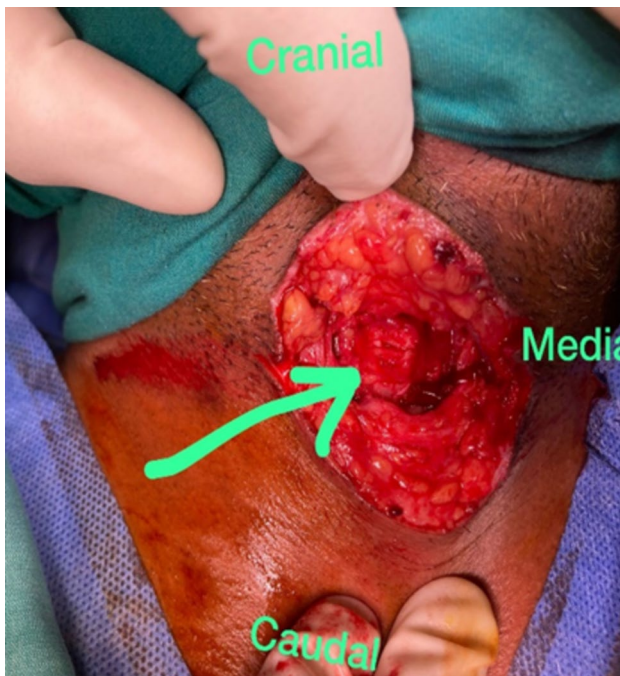


Fig. 8 Thyroid cartilage laceration

The peak age of occurrence was noted in the age group of 20–45 years. In comparing retrospective vs prospective cohort data there were no significant differences noted in mean age and gender involvement. Our result was in concordance with most studies. Al-Thani et al. reported the mean age was 31 ± 9 years and a majority (98%) was males (3). Zakaria et al. too have reported that in patients with TNI males outnumbered females (i.e., 87.4% vs 13.6%) and the average age was 36 years (5).

In the present study, the most common type of injury was penetrating injury noted in 91.0% cases while blunt injury was noted in only 9.0% cases. This observation of the study was in concordance with the result reported by Irish et al. in their study where 78% patients had a penetrating neck injury while 22% patients had blunt neck injury (6). In contradictory to this, Farner et al. have reported that the majority of patients with TNI had sustained blunt neck injuries (65.5%) (7).

Studies have highlighted that blunt neck injury is about 5% of all traumatic neck injuries, mainly following motor vehicle crash. The kinematics of the injury is due to the sudden acceleration-deceleration mechanism which results in the crushing of the trachea or compression of the oesophagus against the cervical spine (3). In our study, we observed blunt neck injury was noted in about 9% of all traumatic neck injuries and in 43% cases it was following RTI. The possible reason could be the increased number of trauma cases following MVC due to RTI. Majority of (90%) road fatalities worldwide are from low and middle-income countries. Pedestrians, cyclists, and motorcycle riders are considered "vulnerable road users" and account for 50% of all road fatalities worldwide. There is a need of strict legislation in India to address all five behavioural risk factors (speed, drink-driving, helmets, seat-belts, and child restraints) and decrease the risk of trauma following RTI (8).

In the present study, the predominant mechanism in TNI was an assault in 32.7% patients followed by in decreasing order RTI in 24.4%, self-inflicted injury in 21.1%, accidental in 12.2%, manja injury in 2.6%. High velocity gunshot injury was noted in 2.6% cases. However, when the data from the both the cohort was analysed, it was noted that in the retrospective cohort the most common mechanism was assault followed by road traffic injury and self-inflicted injury, but in prospective data analysis, road traffic injury was observed as the most common cause followed by assault and self-inflicted injury at the same frequency. This result was in concordance with the result reported by Zakaria et al. in their study where assault was the most common cause of injury in 55.2% cases, followed by deliberate self-harm (DSH) in 38% and accidental injury in 6.8% cases (5).

A zone I and zone III incidence in neck injuries were 13.5 and 1%, respectively in this study, as opposed to 15 and 24% and 14 and 18% in previously published data (9–11).

These variations might be explained by a lower incidence of ballistic injuries (high-velocity injuries) in our study population. Because of their anatomical protection, zones I and III may be less prone to the low-velocity penetrating injury too (Table 7 and Figs. 3, 4, 5, 6, 7, 8).

Strengths and Limitations of the Study

- This was a large ambi-spective study done at a level 1 trauma centre with an adequate sample size.
- We do want to acknowledge the limitations of the study. This is a single-centre observational study with inherent bias such as selection bias and recall bias.
- In addition, there was lack of long-term follow-up for patients with TNI (Traumatic neck injury).

Conclusion

PNI (penetrating neck injury) was the most common injury pattern predominantly following assault or self-inflicted harm. But in the recent times there was a change in the trend toward blunt neck injury pattern. Patients with TNI (Traumatic neck injury) who were hemodynamically stable at the time of initial presentation had good outcome if diagnosed and managed timely.

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